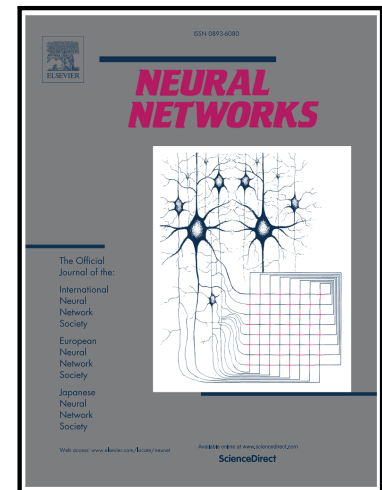


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DeepJSCC for Video Semantic Communication with General Semantic Preservation

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Abstract

Video-based intelligent applications such as autonomous driving, video understanding, and telemedicine rely heavily on the availability of robust and transferable semantic representations. In practical systems, however, video signals transmitted over wireless links are often exposed to severe and unpredictable distortions caused by bandwidth limitations, time-varying noise, and transmission impairments. Under such conditions, high reconstruction quality does not necessarily guarantee the preservation of semantic structures that support downstream reasoning, leading to degraded temporal coherence and poor generalization to unseen tasks. Existing learning-based video transmission and compression methods predominantly optimize pixel-level reconstruction fidelity, while the preservation of general, task-agnostic semantic representations remains insufficiently explored. To address this challenge, we propose a novel end-to-end deep joint source-channel coding (DeepJSCC) framework for video semantic transmission, which explicitly preserves general semantic information for diverse downstream video tasks over wireless channels. Our framework firstly employs a Video-level Semantic Alignment Module (VSAM) to reduce semantic distance between original and reconstructed videos by video-level contrastive learning objectives. Additionally, we propose the Temporal Consistency Learning Module (TCLM) to enhance temporal coherence by predicting future-frame semantic embeddings,

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enabling more stable temporal dynamics in reconstructed videos. The experiments demonstrate that the proposed framework achieves reconstruction quality comparable to existing DeepJSCC-based and traditional transmission schemes, while delivering consistent performance gains across multiple video downstream tasks without any task-specific fine-tuning.

Keywords: DeepJSCC, Video transmission, semantic communication, general semantics

1. Introduction

As video-based intelligent applications such as autonomous driving, video understanding, and telemedicine continue to proliferate, wireless video transmission is increasingly required to support not only high-quality visual reconstruction but also reliable semantic understanding [1]. Traditional communication systems focus on reconstructing the bit stream at the receiver that is consistent with the transmitter to ensure accurate restoration of bit-level content, without explicitly considering the semantic usability of reconstructed videos for artificial intelligence applications. Consequently, the reconstructed videos tend to exhibit weakened semantic representations and degraded temporal coherence, leading to poor performance in downstream tasks that rely on consistent and transferable video semantics [2]. This mismatch highlights the limitations of conventional video transmission schemes in supporting intelligent applications that require robust and general semantic information rather than purely visual reconstruction quality.

Different from traditional communication paradigms, semantic communication [3] focuses on the essence of information delivery, paying attention to whether the semantic content of information can be accurately conveyed, rather than just accurate restoration at the bit level. In semantic communication systems, the receiver places greater emphasis on whether it can successfully restore the semantic-level information or complete specific intelligent tasks [4]. Recent studies have further extended semantic communication to interactive multimedia scenarios, such as real-time video conferencing and cross-modal haptic interaction, showing its potential in practical intelligent communication systems [5, 6]. Among existing approaches, Deep Joint Source-Channel Coding (DeepJSCC) has emerged as a key technique for semantic communication. It uses artificial neural networks (ANNs) to achieve joint optimization of source representation and channel transmission [7], map-

ping source data into continuous-valued channel symbols and effectively optimizing transmission efficiency during the communication process. Recent advances in the field of image semantic communication have shown that methods based on DeepJSCC not only maintain high reconstruction quality but also demonstrate superior performance to traditional methods in specific intelligent tasks (e.g., classification and detection) [8, 9]. In contrast, research on DeepJSCC for video transmission remains limited. Most existing methods draw on the paradigm of video compression and are mainly divided into residual coding [10, 11] and conditional coding [12]-[15]. For example, Deepwive [16] combines bidirectional optical flow estimation with reinforcement learning-driven dynamic bandwidth allocation strategy, aiming to maximize visual reconstruction quality under different bandwidth constraints. MDVSC [17] achieves efficient video transmission under low bandwidth by extracting common semantic features across video frames. DVST [18] uses nonlinear transformation modules and conditional coding architecture to adaptively extract semantic features across video frames.

However, these methods mainly focus on reconstruction quality and transmission efficiency, while neglecting the preservation of general semantic information, resulting in poor performance on downstream tasks. Unlike task-specific semantics tailored to a single application, general semantic information in this paper refers to task-agnostic and transferable semantic representations that are not optimized for a specific downstream task and are expected to remain useful across different downstream video analysis tasks. Explicitly or implicitly retaining this type of information is critical to the usability of videos for downstream tasks such as autonomous driving, telemedicine, and video understanding.

Currently, self-supervised learning and video representation methods [19]-[26] have demonstrated the potential of general semantic features in multi-task transfer. However, research on general video semantics and transmission remains limited. [27] decomposes input images into texture representations and semantic maps and compresses them into the bitstream, balancing reconstruction quality and semantic usability at low bit rates; [30] proposed semantic contrastive coding, which achieves a trade-off between image reconstruction quality and downstream inference performance by jointly optimizing the contrastive loss and reconstruction loss. For video sources, recent studies have explored codec-assisted semantic coding frameworks for video semantic compression. Recent works [28, 29] combine traditional video codecs with neural networks and use masked patch prediction mechanisms

to learn semantic representations. In a similar direction, the low-bitrate video understanding framework in [52] employs a hybrid traditional-neural coding pipeline, where a learned semantic stream is directly encoded by the neural branch. This framework is optimized with a bottleneck-map-based contrastive learning objective, where the video features used for contrastive learning are obtained through bottleneck-map extraction and temporal aggregation. Free-VSC in [53] proposes an unsupervised video semantic compression framework guided by visual foundation models, which uses foundation-model-based semantic alignment and a dynamic trajectory-based entropy model for video semantic compression. In addition, EAN [54] proposes an event-adaptive action recognition network to model video temporal dynamics and short-term motion cues. Although these studies provide useful insights into semantic compression and temporal representation learning, they are mainly built upon source-coding or codec-assisted compression pipelines, or focus on task-oriented video representation learning. Different from these methods, our work studies video semantic transmission and explicitly targets the preservation of general semantic information in an end-to-end DeepJSCC framework. Moreover, instead of only learning semantic representations before compression, we impose semantic alignment and temporal consistency constraints on reconstructed videos after channel transmission, so that transferable video-level semantics can be better preserved under wireless distortion. Our goal is to ensure that reconstructed videos can effectively support diverse downstream video analysis tasks without task-specific fine-tuning, exhibiting better adaptability and robustness in multi-task transfer.

To this end, we propose a novel end-to-end DeepJSCC-based video semantic transmission architecture that integrates contrastive learning into video semantic communication. Firstly, addressing the issue that existing methods mainly optimize reconstruction quality and transmission costs while neglecting general semantic information, we propose the Video-level Semantic Alignment Module (VSAM) that leverages contrastive learning to reduce the semantic distance between the original and the reconstructed videos while maintaining the distinguishability between different videos, thereby explicitly enhancing the ability to preserve general semantics. Then, to mitigate the interference caused by redundant non-semantic information, we propose a frame-level feature aggregation method based on a lightweight binary graph extraction module to encode semantic relationships with low computational and transmission overhead, which helps suppress semantically irrelevant information. Furthermore, considering the dependence of video tasks on tem-

poral consistency, we also propose the Temporal Consistency Learning Module (TCLM) to strengthen the modeling of the global temporal dimension and improve the usability of reconstructed videos in downstream intelligent tasks. Finally, faced with the contradiction between reconstruction quality and downstream task performance, we propose a multi-stage training strategy to gradually guide the model to achieve a good balance between different objectives. It is important to emphasize that our method is not trained or fine-tuned for a specific downstream task, but rather aims to preserve general semantic information, ensuring that the reconstructed videos have good adaptability and robustness in unknown downstream tasks and complex channel environments.

The main contributions of this paper are as follows:

1. We propose a novel end-to-end DeepJSCC-based video semantic transmission architecture that explicitly preserves general semantic information. Our architecture maintains high reconstruction quality in complex channel environments while remaining adaptable to diverse downstream tasks without requiring fine-tuning for specific downstream tasks.
2. We introduce the Video-level Semantic Alignment Module (VSAM), which incorporates contrastive learning into video semantic communication and enhances the preservation of general semantics and improves the robustness of reconstructed videos to diverse downstream tasks. Also, we combine intra-frame and inter-frame dual-layer contrast constraints to further enhance the general semantic consistency and temporal alignment capabilities of reconstructed videos.
3. We propose the Temporal Consistency Learning Module (TCLM), which models global temporal dependencies and improves usability in downstream tasks through a future-frame semantic prediction task.
4. Our method maintains reconstruction performance comparable to state-of-the-art DeepJSCC and traditional schemes, while achieving significant improvements in multiple downstream inference tasks.

The rest of this paper is organized as follows. Section 2 introduces the basic DeepJSCC video communication framework and our proposed video semantic communication framework. Section 3 presents the proposed methods and the overall semantic transmission framework. Section 4 describes the experimental setup and discusses the results. Finally, Section 5 concludes the paper.

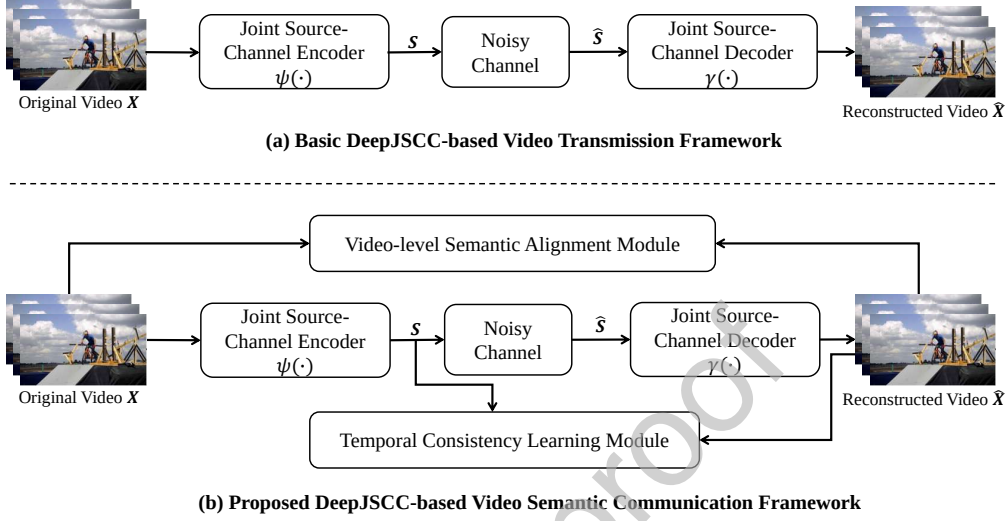


Figure 1: (a) Basic DeepJSCC-based Video Transmission Framework; (b) Proposed DeepJSCC-based Video Semantic Communication Framework.

2. System Model

Considering a wireless video transmission problem with bandwidth and power constraints, we first introduce basic DeepJSCC-based video transmission framework, as illustrated in Fig. 1 (a). Given the group of pictures (GOP) of video sequence $\mathbf{X} = \{\mathbf{x}_t\}_{t=1}^n$ including n frames, where each frame $\mathbf{x}_t \in \mathbb{R}^{w \times h \times 3}$ is represented as a vector of 24-bit RGB image. Herein, w and h denote the width and height of the video frame, respectively. Each GOP is typically composed of one intra-coded frame (I-frame) and $n - 1$ predictive coded frames (P-frames). The I-frame provides a full reference of scene content, while P-frames are temporally correlated with preceding frames and capture inter-frame motion information. Since the transmission of I-frame can be naturally handled using existing image transmission schemes, in this work we mainly focus on the transmission of P-frames, where temporal dependencies and motion information play a critical role.

At the transmitter, the joint source-channel encoder $\Psi : \mathbb{R}^{n \times w \times h \times 3} \rightarrow \mathbb{C}^{\sum_{t=1}^n m_t}$, maps the original video source $\mathbf{X}$ to a sequence of complex-valued channel symbols $\mathbf{S} = \{\mathbf{s}_t\}_{t=1}^n$, $\mathbf{s}_t \in \mathbb{C}^{m_t}$, where $\mathbf{s}_t$ denotes the encoded representations of the current frame $\mathbf{x}_t$ with dimension m_t . After trans-

mission through the channel, the received symbols are denoted as $\hat{\mathbf{S}}$. The decoder then reconstructs the source $\hat{\mathbf{X}} = \Upsilon(\hat{\mathbf{S}})$ by the decoding function $\Upsilon : \mathbb{C}^{\sum_{t=1}^n m_t} \rightarrow \mathbb{R}^{n \times w \times h \times 3}$. Typically, $m_t < w \times h \times 3$, $t \in [1, n]$, which stands for bandwidth compression. We define the channel bandwidth ratio as

$$R = \frac{\sum_{t=1}^n m_t}{n \times w \times h \times 3}, \quad (1)$$

which denotes the average bandwidth required to transmit one pixel. For the channel model, we mainly consider an additive white Gaussian noise (AWGN) channel, formulated as

$$\hat{\mathbf{S}} = \mathbf{S} + \mathbf{N}, \quad \mathbf{N} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}), \quad (2)$$

where $\mathbf{N}$ denotes samples following independent Gaussian distribution with zero mean μ and covariance $\sigma^2 \mathbf{I}$ ($\mathbf{I}$ is the identity matrix). The channel signal-to-noise ratio (CSNR) is defined as follows:

$$CSNR = 10 \log_{10} \frac{1}{\sigma^2} \text{ (dB)}. \quad (3)$$

Meanwhile, we impose an average power constraint at the transmitter:

$$\frac{1}{m_t} \mathbb{E}[\|\mathbf{s}_t\|_2^2] \leq P. \quad (4)$$

Different from typical DeepJSCC video transmission methods, our proposed framework explicitly introduces two core modules into the end-to-end DeepJSCC video transmission structure. As shown in Fig. 1 (b), the first module is the Video-level Semantic Alignment Module (VSAM), which enhances the consistency of general semantics by minimizing the semantic distance between the original and reconstructed videos; the second one is the Temporal Consistency Learning Module (TCLM), which enhances the global temporal modeling capability across frames by utilizing future frame predictions. These improvements ensure that the system not only maintains high reconstruction quality under constrained bandwidth and complex channels but also effectively preserves semantic information useful for downstream intelligent tasks, thereby achieving greater adaptability and robustness. The details of the proposed core modules will be described in detail in Section 3.

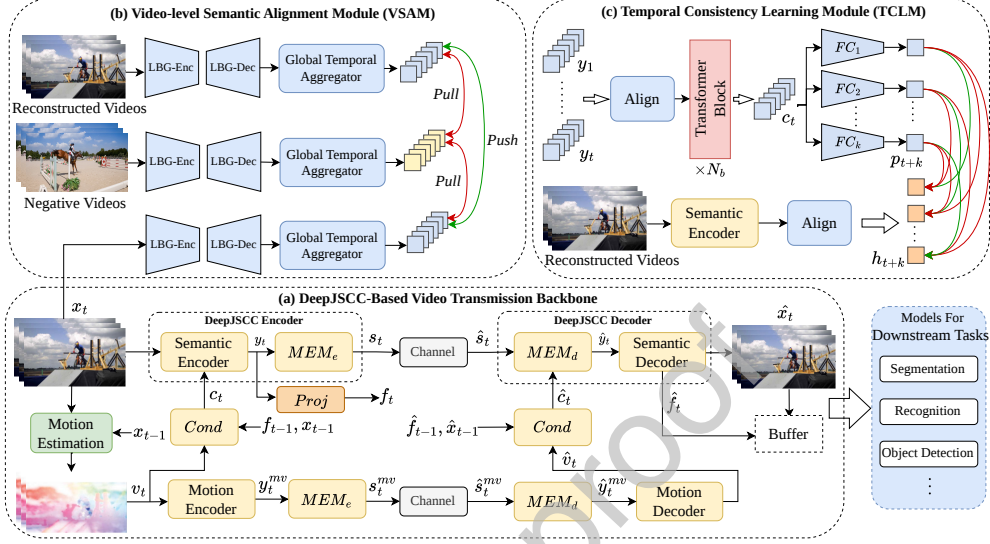


Figure 2: Overview of the proposed framework. We designed two learning modules to help reconstruct videos that retain more general semantic information: (a) The Video-level Semantic Alignment Module (VSAM); (b) The Temporal Consistency Learning Module (TCLM). The reconstructed videos can be directly applied to downstream tasks without any fine-tuning.

3. Proposed Method

3.1. Overview

We aim to go beyond traditional goals of visual reconstruction and explicitly address the preservation of general semantic information. This design choice is motivated by the observation that high-quality reconstructions do not necessarily guarantee usability in downstream tasks, especially under constrained wireless channels. To this end, we build our video semantic communication framework upon the DeepJSCC-based video transmission backbone proposed in [31], and further introduce the Video-level Semantic Alignment Module (VSAM) and the Temporal Consistency Learning Module (TCLM), as illustrated in Fig. 2, enabling reconstructed videos to retain both visual fidelity and semantic usability.

The transmission backbone follows the DeepJSCC-based video transmission framework in [31], which adopts a conditional coding paradigm and consists of Semantic Feature Encoder/Decoder pairs (f_e, f_d), Motion Encoder/Decoder pairs (f_e^{mv}, f_d^{mv}), Motion Estimation (ME) network, feature

projector *Proj*, a condition generation network (*Cond*), Mask-based Entropy Module with encoder and decoder components (MEM_e, MEM_d).

1) Motion Vector Transmission: The input frame $\mathbf{x}_t$ and the previous frame $\mathbf{x}_{t-1}$ are input into *ME* to generate the optical flow $\mathbf{v}_t$:

$$\mathbf{v}_t = ME(\mathbf{x}_t, \mathbf{x}_{t-1}). \quad (5)$$

The *ME* uses an optical flow estimation network [32]. Then, we use f_e^{mv} to map the $\mathbf{v}_t$ to the latent space and get the latent variable $\mathbf{y}_t^{mv}$. The variable rate transmission is realized by MEM_e . Both MEM_e and MEM_d adopt the mask-based entropy module proposed in [31].

$$\mathbf{s}_t^{mv} = MEM_e(f_e^{mv}(\mathbf{v}_t)), \quad (6)$$

where $\mathbf{s}_t^{mv}$ are the channel input symbols. The outputs $\mathbf{s}_t^{mv}$ of MEM_e are reconstructed into the latent variable $\hat{\mathbf{y}}_t^{mv}$ by MEM_d after passing the wireless channel. Finally, the latent variable $\hat{\mathbf{y}}_t^{mv}$ are decoded by f_d^{mv} to get the decoded motion vector $\hat{\mathbf{v}}_t$.

$$\hat{\mathbf{v}}_t = f_d^{mv}(MEM_d(\hat{\mathbf{s}}_t^{mv})). \quad (7)$$

2) Context Generation: To extract spatio-temporal information from historical frames and enhance the coding efficiency, we adopt contexts including the motion vector $\mathbf{v}_t$, ground-truth previous frame $\mathbf{x}_{t-1}$ and the intermediate features $\mathbf{f}_{t-1}$. Specifically, we first use the motion vector $\mathbf{v}_t$ to align $\mathbf{f}_{t-1}$ with $\mathbf{x}_{t-1}$ and generate multi-scale conditions $\mathbf{c}_t$ by *Cond* in [15]. Then the conditions at the transmitter are input into f_e to mine the reference information provided by the contexts. Similarly, at the receiver, the decoding conditions are generated and input to f_d . Different from the transmitter, the contexts at the receiver consist of the decoded motion vector $\hat{\mathbf{v}}_t$, the reconstructed previous frame $\hat{\mathbf{x}}_{t-1}$ and the decoded intermediate features $\hat{\mathbf{f}}_{t-1}$.

$$\mathbf{c}_t = Cond(\mathbf{v}_t, \mathbf{f}_{t-1}, \mathbf{x}_{t-1}), \quad (8)$$

$$\hat{\mathbf{c}}_t = Cond(\hat{\mathbf{v}}_t, \hat{\mathbf{f}}_{t-1}, \hat{\mathbf{x}}_{t-1}). \quad (9)$$

3) Semantic Feature Transmission: With the assistance of multi-scale temporal conditions $\mathbf{c}_t$, f_e encodes the current frame $\mathbf{x}_t$ into the intermediate feature $\mathbf{y}_t$:

$$\mathbf{y}_t = f_e(\mathbf{x}_t, \mathbf{c}_t). \quad (10)$$

Also, we set up *Proj* at the transmitter to obtain the intermediate features $\mathbf{f}_t$ which are propagated to the encoding process of the next P-frame through generating the conditions together with the motion vector $\mathbf{v}_t$ and the reference frame $\mathbf{x}_{t-1}$. Then MEM_e realizes the variable rate transmission and generate the outputs $\mathbf{s}_t$:

$$\mathbf{s}_t = MEM_e(\mathbf{y}_t). \quad (11)$$

$\mathbf{s}_t$ are transmitted as the channel input and the corresponding decoder receives the channel outputs $\hat{\mathbf{s}}_t$, which firstly undergoes the reconstruction of the latent variable done by MEM_d . The recovered intermediate feature $\hat{\mathbf{f}}_t$ and the reconstructed frame $\hat{\mathbf{x}}_t$ are obtained by f_d :

$$\hat{\mathbf{f}}_t, \hat{\mathbf{x}}_t = f_d(MEM_d(\hat{\mathbf{s}}_t), \hat{\mathbf{c}}_t). \quad (12)$$

Furthermore, the intermediate features $\hat{\mathbf{f}}_t$ are also propagated to the decoding process of the next P-frame.

Beyond the DeepJSCC-based Video Transmission backbone, we introduce two key modules that explicitly address the preservation of general semantic information. The video-level Semantic Alignment Module (VSAM) employs a contrastive objective to bridge the semantic gap between original and reconstructed videos, preserving transferable semantics against loss during compression and transmission. The Temporal Consistency Learning Module (TCLM) employs a future frame prediction objective to encourage the learned representations to capture temporally coherent and stable semantic dynamics across frames, thereby improving their suitability for downstream video analysis tasks. The details of these core components will be described in Section 3.2 and 3.3. With our proposed modules, our framework can reconstruct RGB video frames that are naturally aligned with the image prior, making them task-decoupled inputs readily applicable to diverse downstream tasks. This design eliminates the need for task-specific fine-tuning or adapters, ensuring broad applicability across different video analysis models.

3.2. Video-level Semantic Alignment Module

To explicitly preserve general semantic information in reconstructed videos, we propose the Video-level Semantic Alignment Module (VSAM), inspired by the success of contrastive learning in self-supervised representation learning. Unlike prior works that mainly focus on pixel-level or task-specific objectives, our design incorporates contrastive learning into video DeepJSCC, with the explicit goal of maintaining general semantic information that is transferable

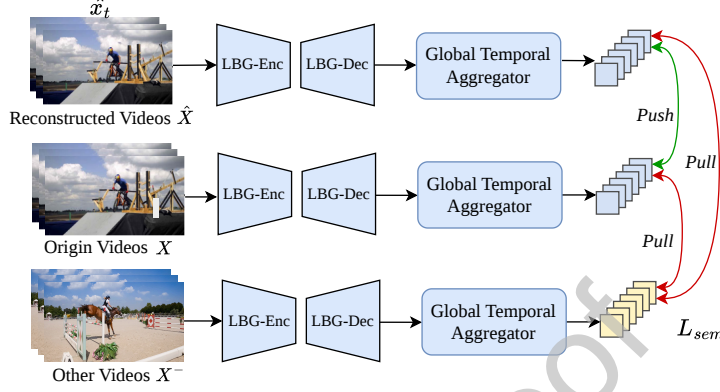


Figure 3: Video-level Semantic Alignment Module (VSAM). For simplicity, we collectively refer to this pair as the Light Binary Graph Module (LBGM), which consists of an encoder (LBG-Enc) and a decoder (LBG-Dec).

across diverse downstream tasks. Specifically, we introduce a video-level contrastive objective that encourages the representations of reconstructed videos to remain close to their original counterparts while remaining distinct from other videos in the semantic space, thereby ensuring the uniqueness of semantic information for each video.

The details of VSAM are shown in Fig. 3. We firstly utilize a Light Binary Graph Module (LBGM) to generate the binary graph of each frame, which suppresses redundant texture details and emphasizes higher-level structural information, such as object shapes and contours. Then, these graphs are input into the Global Temporal Aggregator (GTA) which integrates temporal information to produce compact video-level embeddings. During training, VSAM is jointly optimized with the video transmission backbone in an end-to-end manner.

The learning objective is to explicitly constrain the semantic distance between reconstructed and original videos. In particular, we enforce the embedding of reconstructed video $\hat{X}$ to be closer to its source X than to other unrelated videos in the feature space. The embedding function $F = \text{GTA} \circ \text{LBGM}$ is responsible for projecting videos into a compact semantic space. At the core of this objective, we adopt the InfoNCE function [20] to define video-level contrastive loss:

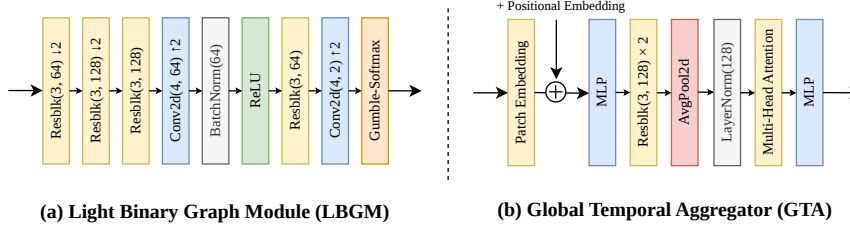


Figure 4: Light Binary Graph Module (LBGM) and Global Temporal Aggregator (GTA). (M, N) represents the kernel size and the number of output channels, and the followed $\uparrow 2/\downarrow 2$ indicates upscaling or downscaling with a stride of 2.

$$\mathcal{L}_{nce} = -\log \frac{\exp \left(F_q(\hat{\mathbf{X}}) \cdot F_k(\mathbf{X}) / \tau \right)}{\sum_{i=1}^N \exp \left(F_q(\hat{\mathbf{X}}) \cdot F_k(\mathbf{X}_i^-) / \tau \right)}, \quad (13)$$

where τ is a temperature parameter, $\mathbf{X}^-$ denotes the negative samples in contrastive learning and N denotes the negative sample number. F_q denotes the query encoder to obtain the video-level representations of the reconstructed anchor videos. F_k denotes the key encoder to obtain the video-level representations of the positive and negative original videos. We use a MoCo-style queue [19] to maintain a large and consistent set of negative video embeddings. The queue stores negative samples and is updated dynamically during training, enabling stable and diverse negative sampling without requiring large batch sizes. For each anchor reconstructed video, all entries in the queue are used as negative video samples, and the current mini-batch is not included in the negative set. We set N to 8192 by default, which is also the queue length.

Light Binary Graph Module (LBGM). The architecture of LBGM consists of an encoder (LBG-Enc) and a decoder (LBG-Dec), as shown in Fig. 4 (a). The encoder consists of two residual blocks with downscaling with a stride of 2. The decoder mainly consists of deconvolution layers and residual blocks, with a Gumbel-Softmax layer at its end to generate the binary map as the frame feature map. Such binary maps highlight high-level semantic structural information (such as object contours and shapes) while suppressing redundant textures, making the features more robust to noise and distortions introduced by wireless transmission.

Global Temporal Aggregator (GTA). The details of GTA are shown in Fig. 4 (b). After obtaining the binary maps from LBGM, each binary map of the frame is divided into non-overlapping 16×16 patches, which are embedded with positional encodings and compressed into 128-dimensional vectors. These patch embeddings are refined through two residual CNN blocks, and then aggregated by average pooling with layer normalization to form a compact frame-level representation. To capture temporal dependencies, the sequence of frame features is fed into a multi-head self-attention module, followed by an MLP projection head that produces the final video-level embedding. Different from masked modeling methods [25] that explicitly suppress non-semantic content by masking spatial regions, the GTA performs implicit semantic selection through temporal correlation modeling. Specifically, the multi-head self-attention operates across frame-level features over time and tends to emphasize features that remain temporally consistent across frames, such as object structure, motion-related patterns, and scene-level semantics. In contrast, semantically irrelevant texture details, such as stochastic noise or unstable fine-grained local textures, are generally less temporally coherent and therefore contribute less during temporal aggregation. In this way, the GTA suppresses non-semantic information without relying on hard masking, which also helps avoid discarding useful semantic information.

To further enhance semantic consistency at different granularities, we further adopt intra-frame and inter-frame contrastive learning strategies, inspired by the design of [33, 22].

Intra-frame contrastive learning. To capture fine-grained appearance consistency, we randomly select one frame $\mathbf{x}_t$ from the current video as the anchor frame, where t is a randomly sampled time index within the current video. We use the reconstructed frame $\hat{\mathbf{x}}_t$ as the positive sample, and other original frames $\mathbf{x}_{t'} (t' \neq t)$ from the same video as the negative samples. The frame-level features corresponding to the sample $\mathbf{x}_t$ are denoted as $\mathbf{u}_t$, which are generated by the GTA module as the embeddings of video frames. For the positive sample $\hat{\mathbf{x}}_t$, the corresponding frame-level feature is denoted as $\hat{\mathbf{u}}_t$, which is extracted from the reconstructed frame. The frame-level features $\mathbf{u}_{t'}$ for negative samples $\mathbf{x}_{t'}$ are extracted from other frames. The loss objective is defined as:

$$L_{intra} = L_{NCE}(g_q(\mathbf{u}_t), \{g_k(\hat{\mathbf{u}}_t)\}, \{g_k(\mathbf{u}_{t'})\}_{t' \neq t}), \quad (14)$$

where g_q and g_k are MLP projection heads for intra-frame contrastive learn-

ing.

Inter-frame contrastive learning. To capture temporal coherence across frames, we extend the contrastive objective from within a frame to across frames. Different from intra-frame contrastive learning, the positives are all the remaining frames $\mathbf{x}_{t'(t' \neq t)}$ from the same video, while negatives $\mathbf{x}_j^-$ are frames sampled from different videos. The corresponding frame-level embeddings are denoted as $\mathbf{u}_{t'}$ and $\mathbf{u}_j^-$, respectively. Hence, the loss objective is defined as:

$$L_{inter} = \frac{1}{|\mathcal{V}|} \sum_{t' \in \mathcal{V}} L_{NCE}(g_q(\mathbf{u}_t), \{g_k(\mathbf{u}_{t'})\}_{t' \neq t}, \{g_k(\mathbf{u}_j^-)\}_{j=1}^N), \quad (15)$$

where $\mathcal{V}$ denotes the set of positive frames from the same video except the anchor frame, and $\{\mathbf{u}_j^-\}_{j=1}^N$ are the corresponding frame-level embeddings of negative samples from other videos. In addition, g_q and g_k here are different from the ones in Eq. (14), but for simplicity of representation, we simplify the notation here.

The overall semantic consistency loss is expressed as:

$$L_{sem} = L_{nce} + \lambda_{intra} L_{intra} + \lambda_{inter} L_{inter}, \quad (16)$$

where λ_{intra} and λ_{inter} balance the contributions of intra-frame and inter-frame contrastive objectives.

3.3. Temporal Consistency Learning Module

In existing video semantic communication systems, the temporal consistency of reconstructed videos is often severely weakened due to the lossy nature of channel and prediction error accumulation. Such temporal degradation typically manifests as inconsistent motion patterns or unstable object dynamics, which directly harms the performance of downstream tasks such as multi-object tracking (MOT) and video object segmentation (VOS), all of which rely heavily on coherent temporal cues.

To explicitly strengthen temporal stability, we propose the Temporal Consistency Learning Module (TCLM) that encourages the transmitted latent representations to encode robust and causally coherent temporal semantics. The key idea is to formulate a future semantic prediction task, where the model predicts the semantic embeddings of future reconstructed frames based on the latent features from past frames. Unlike conventional video prediction methods, our objective is not to synthesize future frames in the pixel

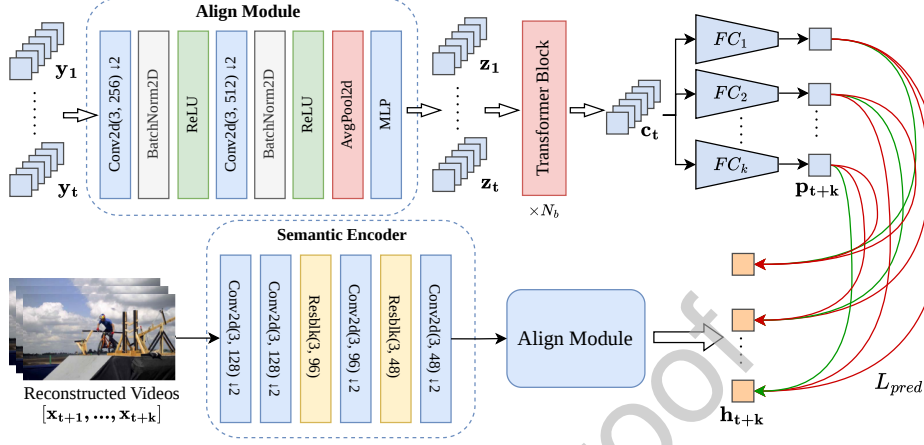


Figure 5: The details of Temporal Consistency Learning Module (TCLM).

domain. Instead, we aim to help the model learn robust representations that preserve meaningful temporal dependencies, thereby improving the temporal consistency of the reconstructed videos.

The details of the Temporal Consistency Learning Module (TCLM) is shown in Fig. 5. Given the latent features $\mathbf{y}_t$, it is aligned by a lightweight align module *Align* to get the aligned features $\mathbf{z}_t$:

$$\mathbf{z}_t = \text{Align}(\mathbf{y}_t). \quad (17)$$

In the last layer of *Align*, we add an MLP head with the output channel of 256. Then, we use a stack of $N_b = 4$ Transformer [47] blocks $F_{context}$ to summarize all $\{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_t\}$ into the temporal context representation $\mathbf{c}_t$ and the process can be described as follows:

$$\mathbf{c}_t = F_{context}(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_t). \quad (18)$$

The context $\mathbf{c}_t$ is used to predict the semantic representation of the next K frames through K parallel fully connected MLP heads FC_i :

$$\mathbf{p}_{t+k} = FC_k(\mathbf{c}_t), k = 1, \dots, K, \quad (19)$$

where $\mathbf{p}_{t+k}$ denotes the predicted results corresponding to the $(t+k)$ th future frame and the output channel of FC_i is set as 256. These predictions are not

compared to original frames, but to the semantic embeddings extracted from reconstructed future frames $\hat{\mathbf{x}}_{t+k}$. We use the Semantic Encoder *SemEnc* and the align module *Align* to extract the semantic embeddings of future reconstructed frames and the process can be described as follows:

$$\mathbf{h}_{t+k} = \text{Align}(\text{SemEnc}(\hat{\mathbf{x}}_{t+k})), \quad (20)$$

where the details of SemEnc are shown in Fig. 5 and it mainly consists of convolution layers and residual blocks.

In order to encourage the model to learn the temporal consistency of videos, we adopt a cross-entropy prediction loss based on cosine similarity. It encourages each predicted embedding $\mathbf{p}_{t+k}$ to match its corresponding target $\mathbf{h}_{t+k}$ while distinguishing it from other samples. The prediction loss is formulated as follows:

$$L_{pred} = -\frac{1}{K} \sum_{k=1}^K \log \frac{\exp(\text{sim}(\mathbf{p}_{t+k}, \mathbf{h}_{t+k}))}{\exp(\text{sim}(\mathbf{p}_{t+k}, \mathbf{h}_{t+k})) + \sum_{\mathbf{h}^- \in N_{t+k}} \exp(\text{sim}(\mathbf{p}_{t+k}, \mathbf{h}^-))}, \quad (21)$$

$$\text{sim}(\mathbf{p}_a, \mathbf{h}_b) = \frac{\mathbf{h}_a(\mathbf{p}_b)^T}{\tau}, \quad (22)$$

where k denotes the future step, $\text{sim}(\cdot)$ denotes cosine similarity, and τ is a temperature hyper-parameter. N_{t+k} denotes a set of negative targets from semantic embeddings from other temporal positions or other videos.

3.4. Optimization Goal

The overall learning goal of our proposed system is to reconstruct videos $\hat{\mathbf{X}}$ that retain rich general semantic information and exhibit great visual quality, while minimizing the bandwidth required for wireless transmission. To achieve this, we design a two-stage training strategy to optimize our proposed system. In the first stage, we train only the DeepJSCC-based video transmission backbone, freezing the parameters of the VSAM and TCLM modules to ensure good visual quality in the reconstructed videos. The training loss in the first training stage can be formulated as follows:

$$L_1 = \frac{1}{n} \sum_{t=1}^n (\lambda R_t + w_t(D_t + D_{f_t})), \quad (23)$$

where R_t denotes the channel bandwidth ratio of the t -th frame as computed in Eq. (1), λ is a balancing weight, and D_t measures the distortion term

between the original frame and the reconstructed frame. We use the Mean Squared Error(MSE) to evaluate the distortion term D_t :

$$D_t = \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2. \quad (24)$$

In addition, we add the feature reconstruction term [31] in our optimization objective which constrains the visual features of the previous reconstructed frame in the pixel domain and enables the network to capture useful information from different contexts. The feature reconstruction loss is defined as:

$$D_{f_t} = \|\mathbf{f}_t - \hat{\mathbf{f}}_t\|_2^2. \quad (25)$$

In the second stage, we jointly train the complete video transmission system based on the first-stage model, including the video-level semantic contrastive learning module and the video temporal consistency learning module. The loss function in the second stage can be formulated as follows:

$$L_2 = \frac{1}{n} \sum_{t=1}^n (\lambda R_t + w_t(D_t + D_{f_t})) + \lambda_{sem} L_{sem} + \lambda_{pred} L_{pred}, \quad (26)$$

where L_{sem} denotes the semantic consistency loss and L_{pred} denotes the temporal consistency loss. The weights λ_{sem} and λ_{pred} control the trade-off between the rate-distortion and semantic preservation. It is worth noting that our optimization framework does not require labels or any task-specific supervision. Instead, the semantic objectives guide the model to preserve more generalizable video semantics that remain beneficial across unknown downstream tasks.

4. Experiments and Results

4.1. Evaluation Datasets

For training, we employ the Vimeo90k [34] datasets on our model and divided the dataset into training and testing sets. All images are randomly cropped to a spatial resolution of 256×256 . For evaluation, we adopt the widely used HEVC benchmark [35] for video coding evaluation. The HEVC dataset consists of 16 YUV420-format videos with various content categories and resolutions, ranging from static backgrounds to high-motion scenarios. This diversity allows us to examine the model's ability to reconstruct videos under different levels of temporal dynamics.

In addition to reconstruction quality evaluation, we further evaluate the usability of the reconstructed videos in downstream intelligent tasks. For video object segmentation (VOS) task, we adopt DAVIS2017 [37], which provides challenging cases of object deformation and occlusion, with all videos from 720p to 1080p. To keep consistency with common VOS settings, we downsampled videos to 480p (854×480) input resolution. For multiple object tracking (MOT) task, we adopt MOT17 [38], which contains complex urban scenes at 1920×1080 resolution. For the video action recognition task, we adopt UCF101 [55] and HMDB51 [56]. UCF101 contains 13,320 video clips from 101 action classes, while HMDB51 contains 6,766 video clips from 51 action categories collected from diverse sources.

4.2. Experimental Setting

Implementation Details. We first trained several models with different values of $\lambda \in \{4e-3, 3e-3, 2e-3, 1e-3\}$ under $CSNR = 10dB$, achieving different bandwidth ratios. And we use the same λ value at all stages of training. In the final training stage, λ_{sem} and λ_{pred} are initialized as 0.1 and 0.05 respectively and are both gradually decayed to 0.001 using a cosine annealing schedule [39]. The negative sample queue length K is empirically set to 8192 and the temperature τ is set to 0.2. For each model, we use the Adam optimizer [40] with a learning rate of $1e-4$ to update the model parameters during training. We used a mini-batch size of 2 and set the GOP size as 4 for training. Also, following [13], we set the loss weights w_t for the four P-frames to (0.5, 1.2, 0.9, 1.2). The system is implemented with PyTorch [41] and it takes about seven days to train the model using one NVIDIA RTX 4090 GPU. All reported experimental results are averaged over multiple independent runs.

As for the downstream task models, we adopt representative baselines across main video understanding tasks. For video object segmentation (VOS), we employ XMem [42], using the official implementation and pretrained weights released by the authors. For multi-object tracking (MOT), we adopt ByteTrack [43], with pretrained models provided by the MMTracking framework [44]. For video action recognition, we employ TSM [57] and SlowFast [58], with pretrained models provided by the MMAAction2 framework [59]. Importantly, all reconstructed videos from different transmission schemes are evaluated using the same pretrained downstream models, and are directly fed into these models without any task-specific fine-tuning.

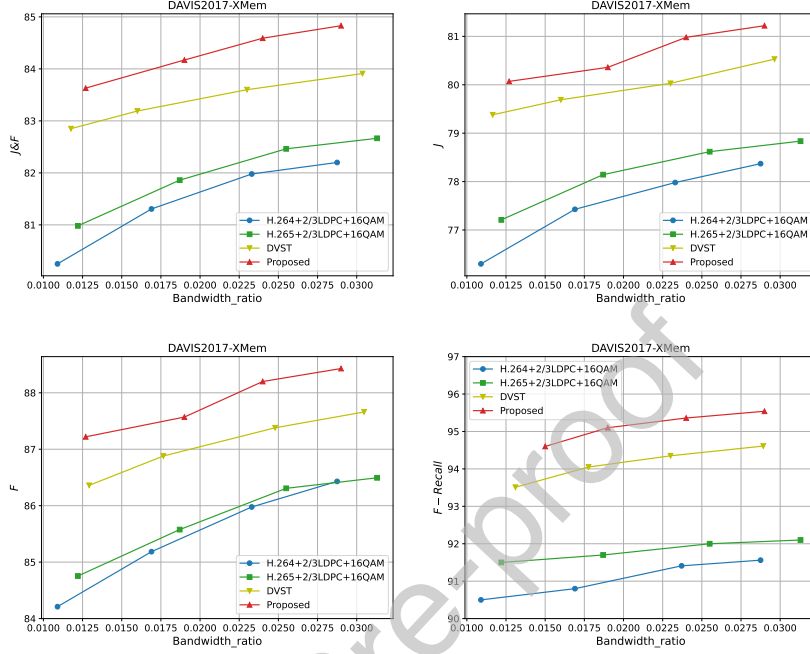


Figure 6: VOS task performance versus the average bandwidth ratio over the AWGN channel (CSNR = 10 dB).

Evaluation Metrics. We utilize the bandwidth compression ratio and CSNR to measure the channel variations. To measure the video reconstruction quality and perceptual quality, we employ the Peak Signal-to-Noise Ratio (PSNR) and Multi-scale Structural Similarity (MS-SSIM) [45]. For the VOS task, we adopt the common DAVIS evaluation protocol, including Jaccard index J , contour accuracy F , their average ($J&F$) and contour recall F -Recall. For the MOT task, multiple complementary metrics are adopted to comprehensively evaluate tracking performance, including Multiple Object Tracking Accuracy (MOTA) [46], which reflects overall tracking accuracy, the detection number of false negatives (FN), which measures missed detections, Identity F1-score (IDF1) [48], which evaluates identity consistency, and Higher Order Tracking Accuracy (HOTA), which jointly assesses detection, association, and localization quality. For the video action recognition task, we use Top-1 Accuracy to evaluate the classification performance.

Baselines and Comparison schemes. We compare our method with both traditional video transmission systems and the recent mainstream video

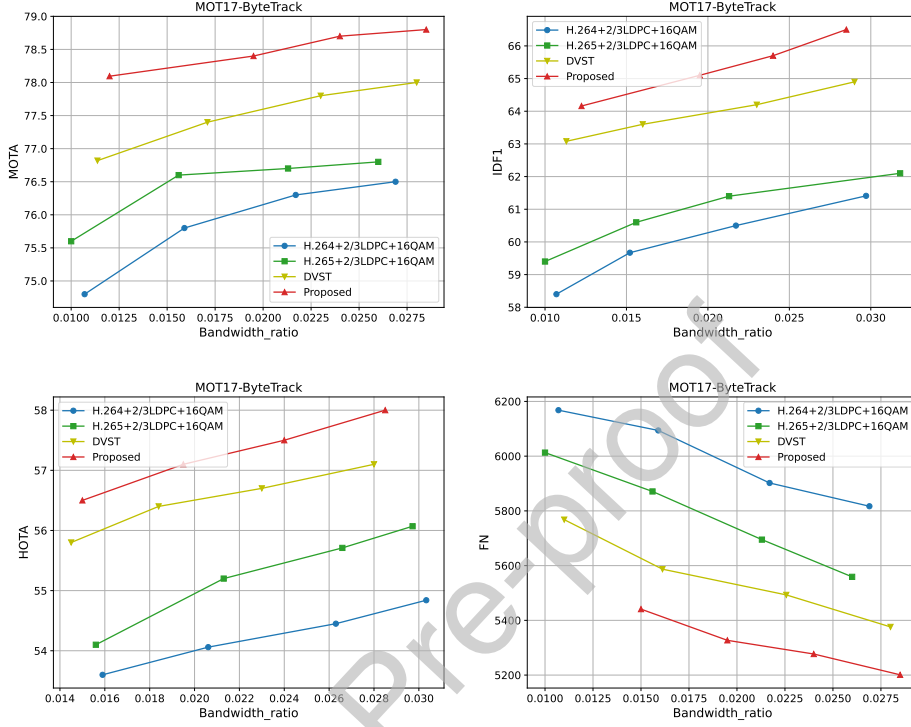


Figure 7: MOT task performance versus the average bandwidth ratio over the AWGN channel (CSNR = 10 dB).

DeepJSCC approach DVST [18]. Specifically, H.264-based and H.265-based transmission schemes are adopted as classical baselines, where source coding is combined with LDPC for channel coding and QAM for modulation. As for the channel model, we consider the widely used AWGN channel, following the same setting as DVST [18]. For H.264 and H.265, we follow the configurations in [18], configuring FFmpeg to the “veryfast” and low-latency mode, with the main difference being that RGB space is used as the input format. Herein, “H.26x+LDPC+QAM” denotes the combination of H.26x, LDPC, and QAM as the entire traditional transmission system compared with our approach. For the traditional transmission schemes, after traversing given combinations of LDPC coded modulation schemes, we finally adopt the configuration with a 2/3 rate (4096, 6144) LDPC code with 16QAM modulation under CSNR = 10dB, which ensures lossless transmission of digital signals.

4.3. Experimental Results

Video object segmentation (VOS). We evaluate different video transmission schemes on the VOS task, which requires both accurate pixel-level details and coherent high-level region semantics to achieve reliable segmentation results. As shown in Fig. 6, our proposed method consistently surpasses the traditional transmission scheme and DeepJSCC-based method DVST [18] across different bandwidth ratios. The superior VOS performance of the proposed method mainly benefits from the explicit modeling of video-level semantics and temporal consistency. VSAM preserves object-level and region-level semantics by aligning original and reconstructed videos in the semantic space, enabling more stable representations. In addition, our proposed TCLM enforces coherent temporal semantics through future-frame prediction, reducing temporal drift and maintaining consistent object appearance across frames. In contrast, traditional schemes and DeepJSCC-based method DVST [18] lack explicit semantic and temporal constraints, leading to degraded usability in fine-grained and temporally sensitive tasks such as video object segmentation.

Multiple object tracking (MOT). We further evaluate different video transmission schemes on the MOT task, which relies on accurate object localization, identity association, and temporally consistent appearance cues to maintain stable tracking results across frames. As shown in Fig. 7, the proposed method consistently outperforms the traditional transmission schemes and DVST [18] across different bandwidth ratios. The results indicate that the reconstructed videos produced by our framework provide more discriminative and temporally stable cues for downstream tracking. This improvement mainly stems from the preservation of consistent semantic representations and temporal continuity in the reconstructed videos, which are essential for identity matching and long-term association. In contrast, traditional schemes and DVST [18] are more susceptible to semantic distortion and temporal inconsistency under bandwidth constraints, leading to degraded tracking robustness.

Video action recognition (VAR). We also evaluate different video transmission schemes on the video action recognition task. As shown in Fig. 8, the proposed method consistently achieves higher Top-1 Accuracy than the traditional transmission schemes and DVST [18] across different bandwidth ratios. This indicates that the reconstructed videos produced by our framework preserve more discriminative action-related semantics, further demonstrating its effectiveness for downstream video understanding tasks.

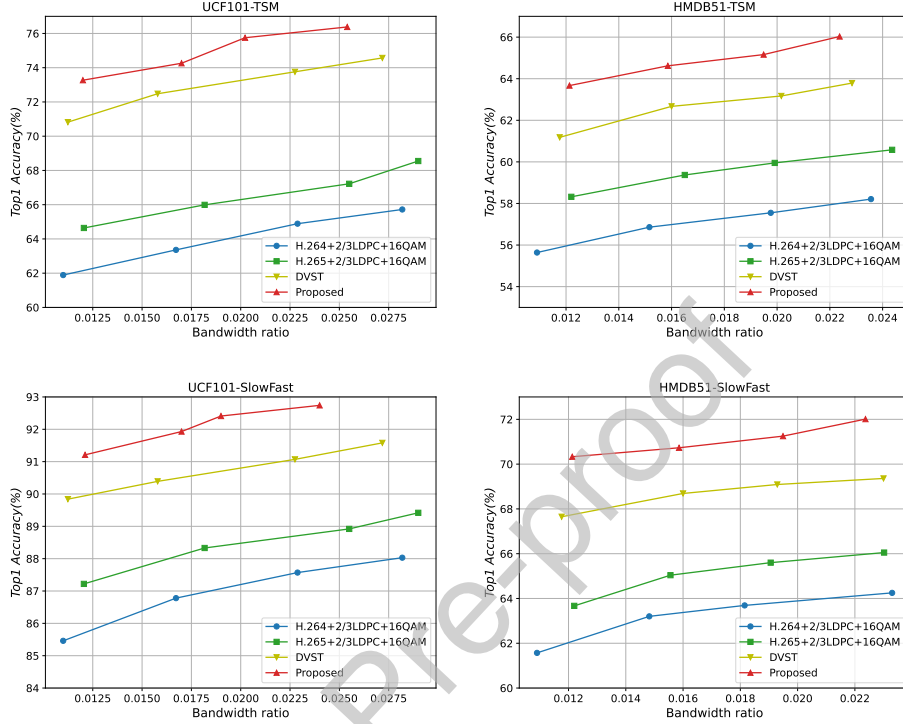


Figure 8: VAR task performance versus the average bandwidth ratio over the AWGN channel (CSNR = 10 dB).

Reconstruction Quality. We further evaluate the reconstruction performance of different video transmission schemes, using PSNR and MS-SSIM as evaluation metrics. As shown in Fig. 9, the proposed method achieves competitive reconstruction quality compared with traditional transmission schemes and DeepJSCC-based method DVST [18] in the HEVC standard dataset. Our proposed framework is further fine-tuned using the rate-distortion loss, which allows the model to balance reconstruction fidelity and transmission efficiency after semantic-oriented representation learning. In terms of PSNR, our proposed framework outperforms the DeepJSCC method DVST [18] and traditional schemes, indicating that the explicit preservation of general semantic information does not compromise reconstruction fidelity. More importantly, when measured by the MS-SSIM metric, the proposed method exhibits clear advantages, especially on high-resolution sequences such as HEVC Class B and Class E under low channel bandwidth ratios.

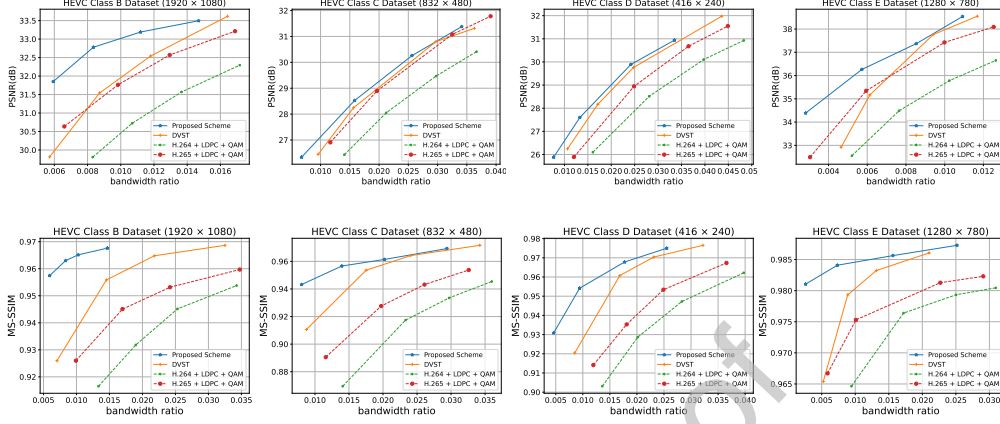


Figure 9: PSNR and MS-SSIM performance versus the average bandwidth ratio over the AWGN channel (CSNR = 10 dB).

This suggests that perceptually important structural information is better preserved when transmission resources are limited. Also, the superior MS-SSIM performance at low bandwidth ratio can be attributed to the semantic-aware design of the proposed framework. By explicitly incorporating general semantic structures into the representation learning process, the proposed framework encourages the preservation of global structures and perceptual consistency, which are particularly critical for high-resolution content. In contrast, traditional schemes and DeepJSCC-based method DVST [18] tend to suffer from structural degradation under aggressive compression, leading to inferior perceptual quality despite comparable PSNR values. Overall, these results demonstrate that the proposed method achieves a favorable balance between reconstruction fidelity and semantic preservation, providing visually consistent reconstructions that are better suited for subsequent semantic analysis tasks.

Visualization Results. Fig. 10 illustrates qualitative reconstruction results on representative samples from the HEVC Class B dataset, including the original frame, our proposed scheme, and traditional transmission schemes. As shown in the cropped regions, it can be observed that the proposed method better preserves structural details and texture continuity, while reducing blocking and ringing artifacts. In contrast, the traditional schemes exhibit more severe structural degradation and loss of fine details, resulting in visually fragmented regions. Overall, the proposed framework provides improved perceptual quality and structural consistency.

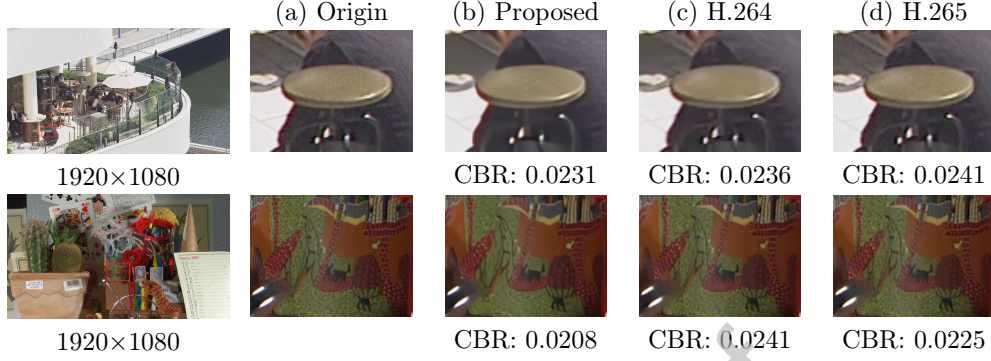


Figure 10: Examples of visual comparison. The first column shows the original frame. In the original frame, we mark the crop image with the red box. The last three columns show the reconstructed frames by using different transmission schemes over the AWGN channel at CSNR = 10 dB.

Fig. 11 further presents qualitative results on the VOS and MOT tasks to illustrate the semantic usability of the reconstructed videos. For the VOS task, compared with the traditional transmission schemes, the proposed method enables more complete and coherent object segmentation results when the reconstructed videos are fed into the same segmentation model. The preserved visual semantics allow the model to better maintain object boundaries and fine-grained details, such as thin structures and object parts (e.g., bicycle wheels), whereas the traditional schemes leads to fragmented masks and missing regions. For the MOT task, we visualize representative tracking results under challenging scenarios. Benefiting from more consistent semantic representations in the reconstructed videos, the proposed method successfully detects and tracks targets with appearances similar to the background, while the traditional schemes fail to reliably identify these objects. These qualitative results further confirm that preserving general semantic and temporal information during wireless transmission is crucial for supporting fine-grained and temporally sensitive video downstream tasks.

To provide more intuitive evidence for the semantic selection effect of LBGM, we visualize the original frame, the reconstructed frame after channel transmission, and the corresponding LBGM binary map, as shown in Fig. 12. Since LBGM is applied at the receiver side, the binary map is generated from the reconstructed frame rather than transmitted through the channel. Although some noisy points appear in the binary map due to chan-

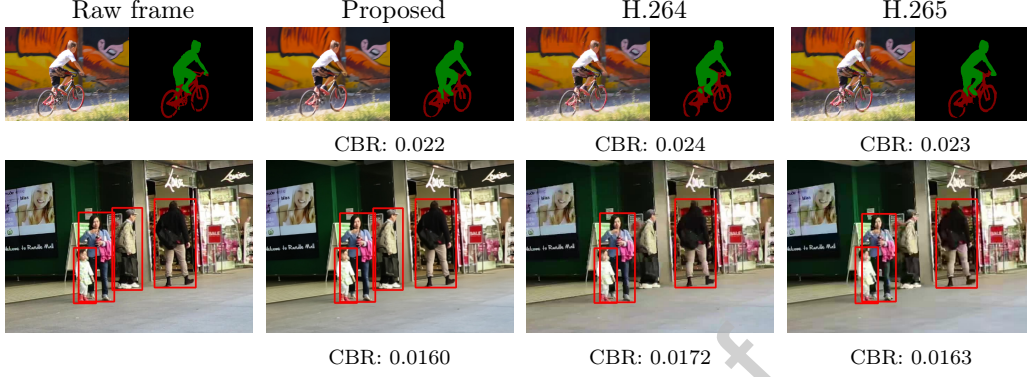


Figure 11: Visualization of the VOS and MOT task results under different transmission schemes over the AWGN channel (CSNR = 10 dB). The first row shows results on the *DAVIS2017-bmxtrees* sequence, and the second row shows results on the *MOT17-09-FRCNN* sequence.



Figure 12: Visualization of the LBGm binary map. From left to right: original frame, reconstructed frame, and receiver-side LBGm binary map.

nel distortion, the main semantic structures, such as the object contour and action-related regions, are still well preserved. In contrast, large background regions and unstable texture details are largely suppressed. This indicates that LBGm can retain semantic-related information while reducing the impact of semantically irrelevant details.

Channel Robustness Analysis. We further evaluate the performance of different video transmission schemes under varying CSNRs to evaluate their robustness to channel conditions. To ensure a fair comparison, we adjusted the bandwidth of the comparison schemes accordingly to ensure that the transmission bandwidths are as close as possible. Fig. 13 illustrates the PSNR performance on the HEVC Class B sequence as CSNR increases. For the comparison schemes, we used combinations of 2/3 LDPC and 4QAM, 16QAM, 64QAM. The proposed scheme exhibits stable performance gains with increasing CSNR and consistently outperforms both traditional trans-

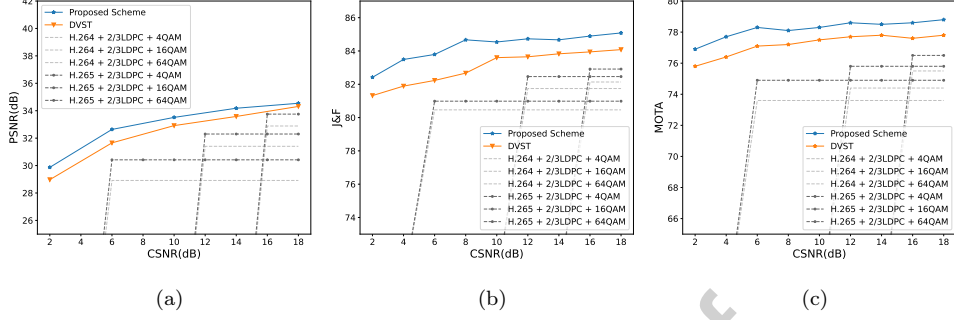


Figure 13: Reconstruction and downstream task performance of relevant schemes under varying CSNRs over the AWGN channel.

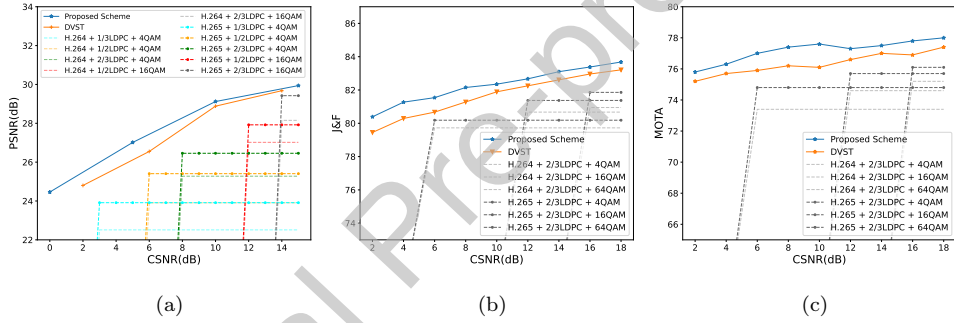


Figure 14: Reconstruction and downstream task performance of relevant schemes under varying CSNRs over the Rayleigh fading channel.

mission schemes and existing DeepJSCC-based method DVST [18] across a wide range of channel conditions. Beyond reconstruction quality, we also analyze the impact of channel conditions on downstream task performance following the same task evaluation protocols and datasets as in the VOS and MOT experiments described above. As shown in Fig. 13, the proposed method achieves consistently higher $J&F$ mean value on the VOS task and higher MOTA on the MOT task under different CSNRs. Notably, even at low CSNRs, the proposed scheme maintains relatively stable task performance, demonstrating stronger robustness and improved semantic usability for downstream video tasks.

We additionally evaluate the proposed framework under the Rayleigh fading channel, and the corresponding results are shown in Fig. 14. Compared

Table 1: Comparison of computational complexity, inference latency, and model size between DVST and our method.

Method	FLOPS	Params	Inference Latency
DVST	484.60G	13.65M	130ms
Ours (train)	844.31G	24.56M	/
Ours (inference)	277.43G	15.47M	113ms

with the AWGN setting, all methods experience certain performance degradation due to the more challenging multi-path fading effect. Nevertheless, the proposed method still maintains superior reconstruction quality and downstream task performance across different CSNRs. This robustness mainly benefits from the semantic alignment and temporal consistency objectives, which encourage the model to preserve semantically stable and transferable video representations even under channel distortions. These results further demonstrate the effectiveness of the proposed framework in practical wireless video transmission scenarios.

We also acknowledge that burst errors are important in practical wireless video transmission. In this work, we focus on AWGN and Rayleigh fading channels, where the channel distortion is modeled at the symbol level in the end-to-end DeepJSCC framework. Burst-error or packet-loss scenarios usually involve temporally correlated distortions, packetization strategies, interleaving mechanisms, and packet-level error concealment, which introduce additional system assumptions beyond the current setting. A systematic evaluation under burst-error scenarios will be considered as an important direction for future work.

Model Complexity. We evaluate the Floating-point Operations Per Second (FLOPS) and parameters required for DVST [18] and our proposed method. For a fair comparison, the FLOPS and inference latency reported in Table 1 are measured under the same input setting, with a spatial resolution of 256×256 and a GOP size of 4. During training, our model includes all modules, including the Video-level Semantic Alignment Module (VSAM) and the Temporal Consistency Learning Module (TCLM), which increases the computational cost. However, during inference, VSAM and TCLM are removed and only the video transmission backbone is used. As shown in Table 1, under the inference configuration, our method requires slightly more parameters than DVST [18], but the amount of floating-point computation

Table 2: Performance influence of different components of our proposed framework over the AWGN channel with CSNR = 10 dB. Reconstruction quality is evaluated on HEVC Class B, while VOS and MOT performance are evaluated on DAVIS2017 and MOT17, respectively.

+FR Loss	+VSAM	+TCLM	PSNR	MS-SSIM	$J\&F$	MOTA
×	×	×	31.43	0.934	82.83	76.8
✓	×	×	32.20	0.959	83.32	77.3
✓	✓	×	32.71	0.961	84.45	78.3
✓	×	✓	32.36	0.960	84.38	78.0
✓	✓	✓	32.78	0.963	84.88	78.5

is significantly reduced, by about 42%, demonstrating a substantial improvement in inference efficiency. In addition, we report the inference latency measured in milliseconds per frame. Our method achieves an average latency of 113 ms/frame, which is lower than that of DVST, demonstrating improved inference efficiency.

4.4. Ablation Studies

We conduct ablation experiments to evaluate the contribution of the feature reconstruction loss (FR Loss), the Video-level Semantic Alignment Module (VSAM), and the Temporal Consistency Learning Module (TCLM). Following the same evaluation protocols as in the main experiments, reconstruction quality, VOS, and MOT performance are evaluated on their respective datasets. All configurations are evaluated over the AWGN channel with CSNR = 10 dB while maintaining closely channel bandwidth ratios to ensure a fair comparison. The results are reported in Table 2. Introducing the FR Loss mainly improves PSNR and MS-SSIM, indicating that feature-level reconstruction supervision provides a stable baseline visual reconstruction quality. Building upon this baseline, incorporating VSAM leads to consistent performance gains on both VOS and MOT tasks, demonstrating that video-level semantic alignment plays a key role in preserving transferable semantic representations beyond pixel-level fidelity. In contrast, adding TCLM primarily improves the performance of temporally sensitive downstream tasks. This is because future-frame semantic prediction can constrain the temporal evolution of semantic representations in the latent space, encouraging adjacent reconstructed frames to maintain more coherent semantic dynamics. As a result, TCLM helps reduce temporal semantic fluctuations caused

Table 3: Performance influence of different components in VSAM of our proposed framework over the AWGN channel with CSNR = 10 dB.

L_{nce}	L_{inter}	L_{intra}	PSNR	MS-SSIM	$J\&F$	MOTA
✓	×	×	32.94	0.964	84.51	78.2
✓	✓	×	32.75	0.962	84.57	78.3
✓	×	✓	33.02	0.965	84.64	78.2
✓	✓	✓	32.78	0.963	84.88	78.5

by channel distortions and improves the stability of object representations across frames. When all components are jointly applied, the proposed framework achieves the best overall performance across reconstruction metrics and downstream task performance indicators. This suggests that while FR Loss provides necessary reconstruction supervision, the improvements in semantic usability and temporal stability are mainly attributed to our proposed VSAM and TCLM.

We further conduct ablation experiments to analyze the effect of different contrastive learning objectives within the Video-level Semantic Alignment Module (VSAM), including the video-level contrastive loss (L_{nce}), the inter-frame contrastive loss (L_{inter}), and the intra-frame contrastive loss (L_{intra}). As shown in Table 3, when only partial contrastive constraints are applied, such as removing either L_{inter} or L_{intra} , our framework can still achieve competitive reconstruction quality, and in some cases slightly higher PSNR or MS-SSIM. However, these configurations fail to consistently deliver the best performance on downstream VOS and MOT tasks. This indicates that local contrastive constraints alone are insufficient to preserve stable video-level semantic representations. In contrast, when all contrastive objectives are jointly applied, VSAM achieves the best overall performance on downstream task metrics. This demonstrates that video-level contrastive learning, together with complementary inter-frame and intra-frame semantic constraints, plays a crucial role in maintaining coherent and discriminative semantic structures across both spatial and temporal dimensions. Overall, these results validate the design of VSAM and confirm that jointly modeling video-level, inter-frame, and intra-frame semantic relationships is essential for preserving general semantic information that benefits diverse downstream video analysis tasks.

We also investigate the sensitivity of the semantic-related loss weights

Table 4: Sensitivity analysis of the semantic-related loss weights λ_{sem} and λ_{pred} over the AWGN channel with CSNR = 10 dB. Reconstruction quality is evaluated on HEVC Class B, while VOS and MOT performance are evaluated on DAVIS2017 and MOT17, respectively.

λ_{sem}	λ_{pred}	PSNR	MS-SSIM	J&F	MOTA
0	0.05	32.36	0.960	84.38	78.0
0.05	0.05	32.94	0.963	84.30	78.3
0.2	0.05	32.98	0.964	84.59	78.3
0.1	0	32.71	0.961	84.45	78.3
0.1	0.025	32.89	0.963	84.49	77.9
0.1	0.1	32.92	0.963	84.29	78.0
0.1	0.05	32.78	0.963	84.88	78.5

Table 5: Sensitivity analysis of the different negative sample queue length K in the VSAM module under the AWGN channel (CSNR = 10 dB).

K	PSNR	MS-SSIM	J&F	MOTA
2048	32.85	0.961	83.74	77.9
4096	32.86	0.963	84.13	78.2
8192	32.78	0.963	84.88	78.5

λ_{sem} and λ_{pred} . The results are reported in Table 4. It can be observed that varying these weights within a reasonable range does not cause an obvious degradation in PSNR or MS-SSIM. Instead, moderate semantic constraints can slightly improve reconstruction quality. From the perspective of downstream task, jointly considering both λ_{sem} and λ_{pred} generally leads to stronger performance than using either of them alone. This suggests that semantic alignment and temporal consistency play complementary roles in improving semantic usability, while still maintaining competitive reconstruction fidelity.

Table 5 presents the sensitivity analysis of the negative sample queue length K used in the VSAM module. It can be observed that varying the queue length has only a limited influence on reconstruction quality, as PSNR and MS-SSIM remain relatively stable across different settings. In contrast, downstream task performance shows a clearer improvement as the queue length increases. In particular, larger queue lengths lead to better down-

Table 6: Sensitivity analysis of the temperature hyper-parameter τ in the VSAM module under the AWGN channel (CSNR = 10 dB).

τ	PSNR	MS-SSIM	J&F	MOTA
0.1	32.86	0.958	84.15	78.2
0.2	32.78	0.963	84.88	78.5
0.4	32.87	0.961	84.29	78.0

Table 7: Sensitivity analysis of different GOP lengths under the AWGN channel (CSNR = 10 dB).

<i>GOP Length</i>	PSNR	MS-SSIM	J&F	MOTA
4	33.70	0.969	84.57	78.7
8	32.94	0.964	83.86	78.6
12	32.31	0.960	83.54	78.5

stream tasks performance results, indicating that a richer set of negative video samples is beneficial for learning more discriminative and transferable semantic representations. Among these settings, $K = 8192$ achieves the best overall downstream performance, which justifies its use as the default configuration in our experiments.

Table 6 presents the sensitivity analysis of the temperature hyper-parameter τ in the contrastive objective of the VSAM module. Different temperature settings have only a small influence on PSNR and MS-SSIM, indicating relatively stable reconstruction quality. By contrast, downstream task performance is more sensitive to τ , and a moderate temperature achieves better overall results. This suggests that the VSAM module is reasonably robust to the choice of τ within a practical range.

Table 7 presents the impact of different GOP lengths on reconstruction quality and downstream task performance. To ensure a fair comparison, the transmission bandwidth ratios under different GOP settings are adjusted to be as close as possible for each evaluation task. As the GOP length increases, both reconstruction quality and downstream-task performance show a slight degradation. This is because larger GOP lengths introduce longer temporal dependencies and increase the difficulty of preserving stable semantic information across frames. Nevertheless, the performance degradation remains relatively limited, indicating that the proposed framework still maintains

good robustness under longer GOP settings.

5. Conclusion

In this paper, we propose an end-to-end DeepJSCC-based video semantic transmission framework that explicitly targets the preservation of general semantic information for downstream video analysis tasks. Different from conventional video transmission methods that mainly optimize reconstruction fidelity, the proposed framework enhances semantic usability by jointly modeling video-level semantic alignment and temporal consistency in the latent space. Specifically, we introduce a Video-level Semantic Alignment Module (VSAM) based on contrastive learning to align the semantic representations of original and reconstructed videos, thereby preserving transferable and task-agnostic video semantics. Additionally, we also propose a Temporal Consistency Learning Module (TCLM) to enforce coherent temporal semantics through future-frame semantic prediction, which improves the temporal stability of reconstructed videos under channel distortions. Extensive experimental results demonstrate that the proposed framework achieves reconstruction quality comparable to a mainstream DeepJSCC-based method and traditional transmission schemes, while consistently outperforming these baselines on multiple representative downstream video tasks, including video object segmentation, multi-object tracking, and video action recognition, without any task-specific fine-tuning. Furthermore, the proposed method exhibits strong robustness across different channel conditions, highlighting its effectiveness for practical wireless video transmission scenarios. Although the current experimental validation is conducted on representative downstream tasks, the proposed framework is expected to remain beneficial for broader downstream video analysis applications. A more comprehensive validation on a wider range of unseen tasks will be investigated in future work.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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